



HINDUSTHAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

Department of Artificial Intelligence and Data Science

PROJECT TITLE: SKILL GAP ANALYSER

ACADEMIC YEAR: 2025-2026

PROJECT STATEMENT: An ML-based prototype that automatically quantifies and itemises the skill gap between a candidate's resume and a target job description, using taxonomy-based skill extraction and TF-IDF + cosine-similarity matching.

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TECHNOVA 2k26 - PROJECT EXPO

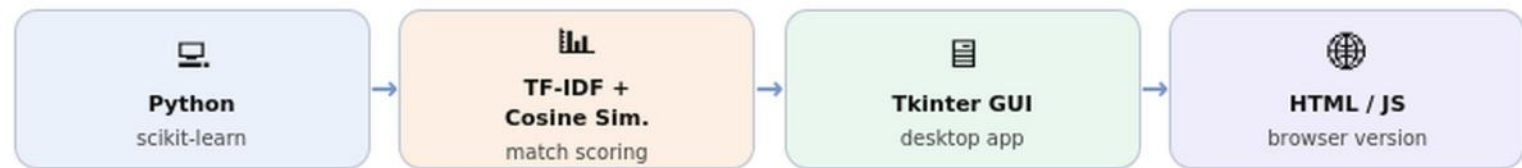
MOTIVATION / OBJECTIVE / INTRODUCTION

IDENTIFY GAPS: Candidates compare themselves to a job title, not its actual skill list, so specific gaps stay invisible. ATS rejections give no explanation of what was missing.

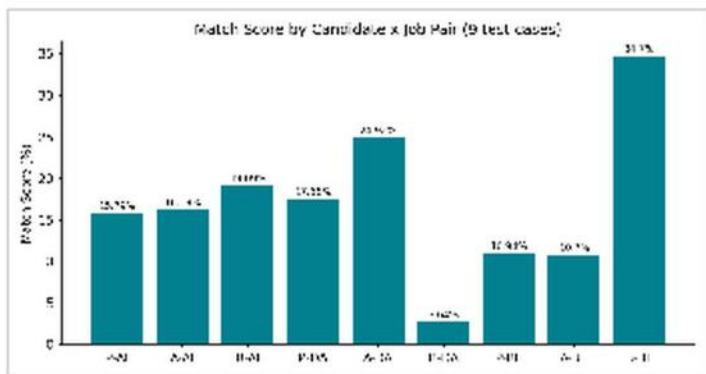
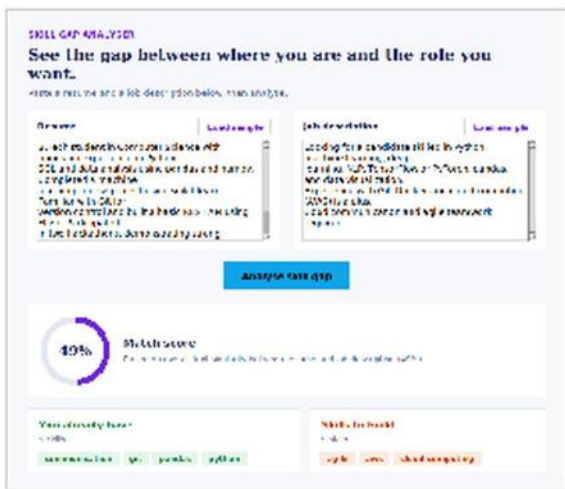
DRIVE INNOVATION: Turn a vague "not a fit" into a precise, itemised list of skills to build — using a transparent, explainable pipeline instead of an opaque black-box score.

ENHANCE PERFORMANCE: Zero-GPU, zero-cloud pipeline averaging 1.57 ms per resume-job pair.

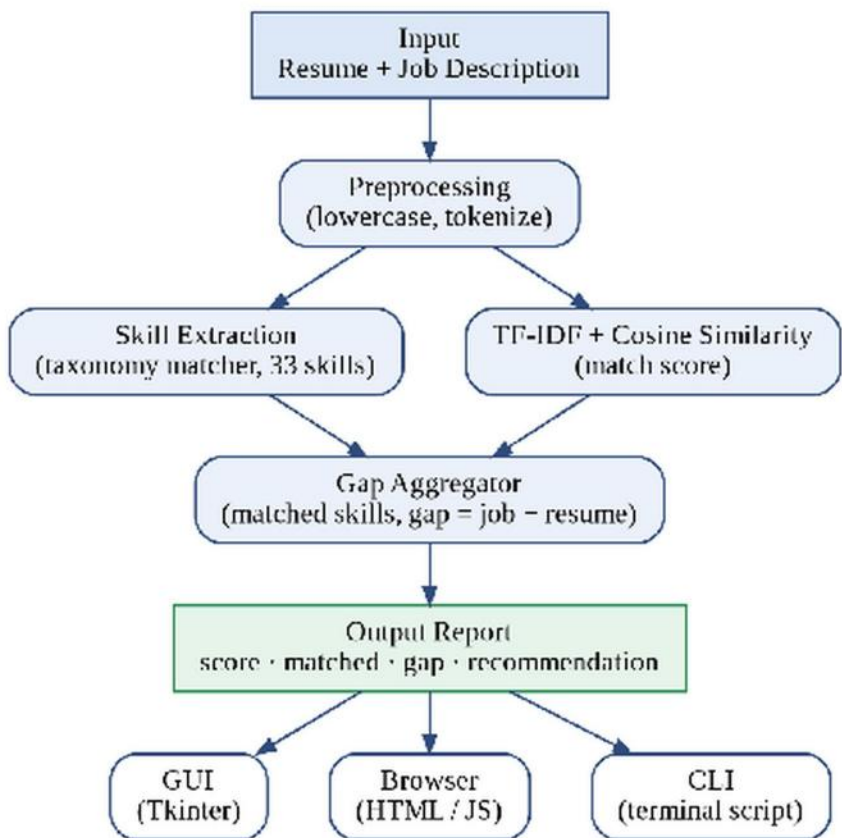
TOOLS & TECHNOLOGY USED



RESULT & OUTPUT



ARCHITECTURE / FRAMEWORK / METHODOLOGY



UNIQUE FEATURE / INNOVATION

NOVEL APPROACH: Deterministic, word-boundary taxonomy matching (33 skills) combined with TF-IDF + cosine similarity — every match is traceable to source text.

USER-CENTRIC FEATURE: Itemised skill-gap report with a direct "what to learn next" recommendation, not a single opaque score.

APPLICATION / USECASE

STUDENTS & JOB SEEKERS: Self-assessment before applying to a role.

PLACEMENT CELLS: Screen many students' readiness for a role at once. **CAREER COUNSELLING:** Concrete, itemised talking points.

CONCLUSION

ACHIEVEMENTS: Validated across 9 resume-job test pairs spanning AI/ML, Data Analyst and Backend Developer roles; delivered consistently through a GUI, browser page and terminal script.

FUTURE ENHANCEMENT: Semantic/embedding-based extraction, a larger dynamic skill taxonomy, resume file upload (PDF/DOCX), and personalised learning-path recommendations.

Three-stage pipeline — Extract → Compare → Recommend — delivered through a Tkinter GUI, browser page, and CLI script.